

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : PEC-CS701E/PEC-IT701D/PECICB702A Machine learning

UPID : 007554

Time Allotted : 3 Hours

Full Marks :70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**1. Answer *any ten* of the following :

[1 x 10 = 10]

- (I) What is unsupervised learning?
- (II) What is an ensemble model in machine learning?
- (III) Define deep learning.
- (IV) Define Reinforcement learning.
- (V) What are support vectors in the context of SVMs?
- (VI) What is PCA?
- (VII) -----machine learning algorithm is based upon the idea of bagging.
- (VIII) Write down the difference between under fitting and over fitting.
- (IX) What is semisupervised learning in machine learning?
- (X) What do you mean by Clustering?
- (XI) Give one example of Dimensionality reduction technique.
- (XII) Give one real life example of deep learning.

Group-B (Short Answer Type Question)Answer *any three* of the following :

[5 x 3 = 15]

2. Explain Precision, Recall and F1 Score. [5]
3. Discuss working of Neural network. State two differences between Deep learning and Machine learning. [5]
4. Define conditional probability. Discuss Bayes theorem with suitable example. [5]
5. Compare linear regression with logistic regression. Write two applications of logistic regression. [5]
6. How does the choice of distance metric affect the performance of a KNN classifier? [5]

Group-C (Long Answer Type Question)Answer *any three* of the following :

[15 x 3 = 45]

7. (a) Define-Support Vectors, Margins and Hyperplane. [6]
 (b) What is the difference between a linear and non-linear SVM? [4]
 (c) Discuss cost function in regression. [5]
8. (a) State the requirement of clustering algorithm? [5]
 (b) How does the clustering differ from classification in machine learning? [5]
 (c) Describe the strengths and weaknesses of the K-Means clustering algorithm. When is it suitable for use? <https://www.makaut.com> [5]
9. (a) Compare between Bagging & Boosting technique. [6]
 (b) State the Difference Between Bias and variance in ML. [5]
 (c) Write short note on Random Forest algorithm. [4]
10. (a) Write Short note on Content Based filtering and Collaborative filtering. [6]
 (b) How does the system make recommendations. [4]
 (c) Discuss mathematics behind the recommendations made using content-based filtering. [5]
11. (a) Discuss Naïve Bayes Classifier algorithm. [4]
 (b) Why it is called Naïve Bayes ? [3]
 (c) Explain KNN algorithm with suitable example. [8]

Using these probabilities estimate the probability values for the new instance (Color=Green,legs =2,Height=Tall,and Smelly=No).

No	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

*** END OF PAPER ***